Web Technologies

Lab 2

Responsive Navbar and Grid Drill

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Program / Semester	BS Computer Science, 5th Semester
Course	Web Technologies
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1. Objective

This lab is a short practice for Assignment 2. The goal was to build a responsive navbar and a responsive product grid and to show how they change on phone, tablet and desktop screens. The content is placeholder text and grey image boxes, because the focus is on the responsive behaviour and not on the final design.

2. What I Built

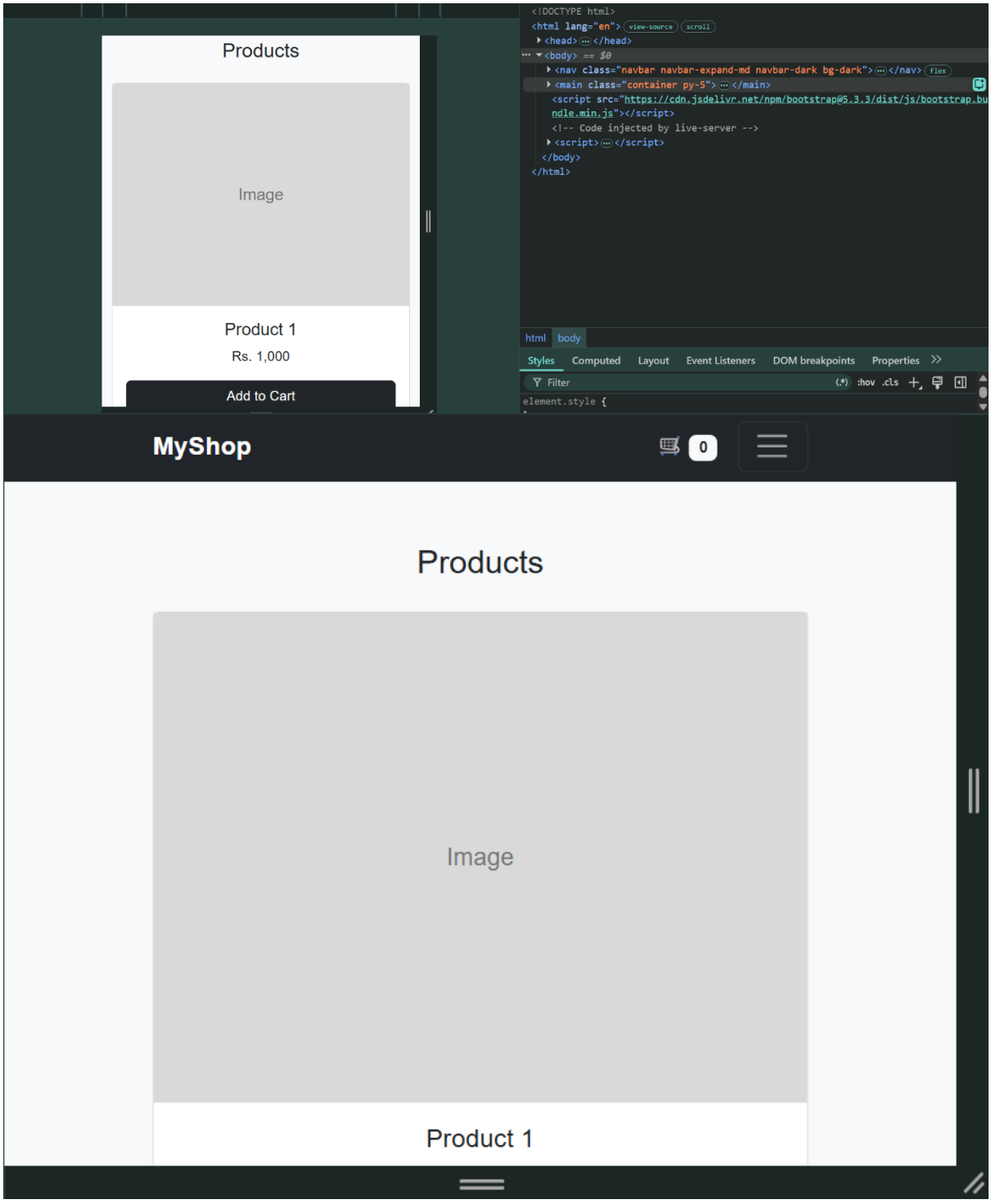
- A navbar with the logo on the left, the links in the center and a cart icon on the right.
- The navbar collapses into a hamburger menu below 768px.
- A product grid of 8 cards with 4 columns on desktop, 2 on tablet and 1 on phone.
- Tools: HTML5, CSS3 and Bootstrap 5 (via CDN), written in Visual Studio Code and tested with Live Server.

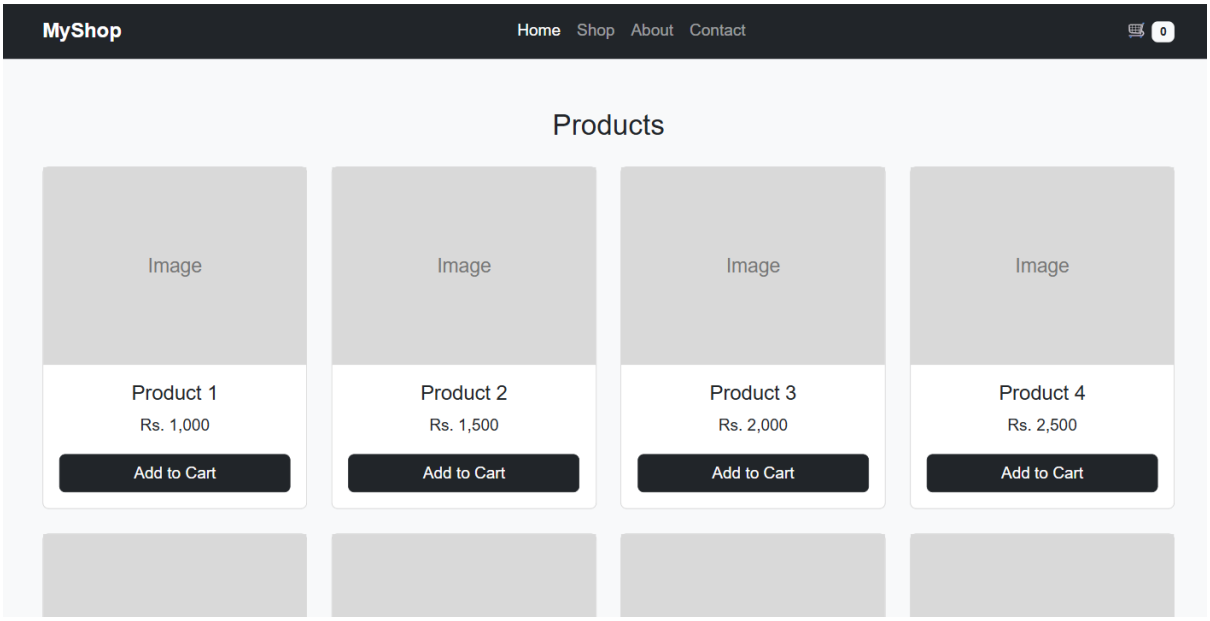
Responsive behaviour

Screen	Width	Columns	Navbar
Phone	Below 768px	1	Hamburger menu
Tablet	768px to 991px	2	Full navbar
Desktop	992px and above	4	Full navbar

3. Screenshots

Screenshots were taken in Chrome DevTools using the device toolbar.


Products
Image
Product 1
Rs. 1,000
Add to Cart
<!DOCTYPE html>
<html lang="en">
<head></head>
<body>
<nav class="navbar navbar-expand-md navbar-dark bg-dark"></nav>
<main class="container py-5"></main>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
<!-- Code injected by live-server -->
<script></script>
</body>
</html>
html
body
Styles
Computed
Layout
Event Listeners
DOM breakpoints
Properties
Filter
element.style {
MyShop
0
Products
Image
Product 1



4. How It Works

Navbar

The navbar uses Bootstrap's navbar component with the class navbar-expand-md. Below 768px the links are hidden inside a collapsible menu and a hamburger button appears. From 768px upwards the links show in a normal bar. The links are centered with mx-auto, and the cart icon is moved to the right with the order-md-last class. The cart icon sits outside the collapsing menu so it stays visible on phones.

Product grid

The grid uses Bootstrap's row-cols classes: row-cols-1 for 1 column on phones, row-cols-md-2 for 2 columns from 768px, and row-cols-lg-4 for 4 columns from 992px. The class g-4 adds the gap between cards, and h-100 makes all the cards in a row the same height.

Flexbox

Bootstrap's grid and navbar are built on Flexbox. I also used Flexbox in my own CSS to center the text inside the grey image boxes (display: flex with align-items and justify-content).

My own CSS

The style.css file sets the grey placeholder image boxes with a fixed shape using aspect-ratio, adds a small hover effect on the cards, and includes one media query that reduces the heading size on phones.

5. Conclusion

This lab gave me practice with the responsive mechanics I will use in Assignment 2: a collapsing navbar and a grid that changes the number of colum